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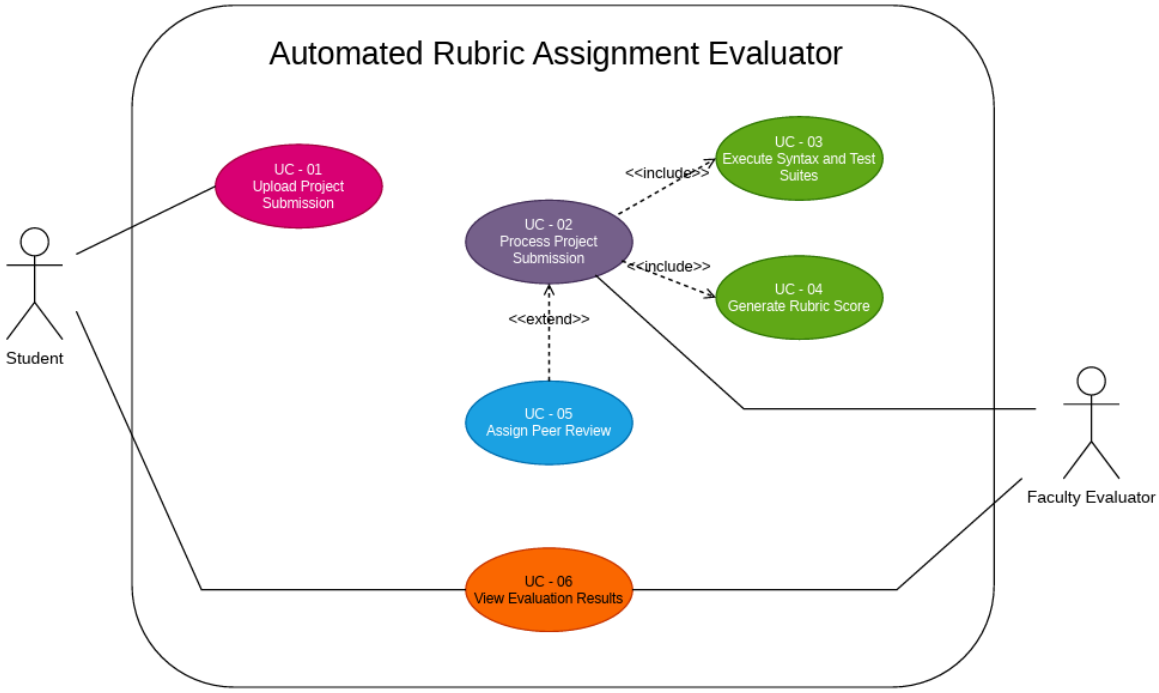
SRN: PES1UG24AM064

Section: B

Requirements Table:

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall automatically queue uploaded student project ZIP files, execute pre-configured unit test suites, and generate an itemized rubric score breakdown.	High	Pass: After a valid project ZIP is submitted, the system queues it, executes the configured tests, and generates an itemized rubric score breakdown. Fail: The submission is not queued, tests are not executed, or the rubric breakdown is not generated.	This is the core automated evaluation workflow described in the problem statement.
FR-002	Functional	The system shall allow students to upload their project submissions as ZIP files for automated evaluation.	High	Pass: A student can select a valid ZIP project file, upload it, and receive confirmation that the submission was received. Fail: The student cannot upload the ZIP file or no submission confirmation is provided.	Project uploads provide the input required for automated evaluation.
FR-003	Functional	The system shall execute the configured syntax checks and unit test suites on submitted project files and record the resulting test outcomes.	High	Pass: For a valid submitted project, the configured syntax checks and unit tests execute and their pass/fail results are recorded. Fail: The checks or tests do not execute or their results are not recorded.	Automated testing is required to evaluate submitted student projects.
FR-004	Functional	The system shall automatically assign submitted projects to students for peer review without requiring manual distribution by faculty.	High	Pass: After a submission is ready for peer review, the system automatically assigns it to an eligible student reviewer and records the assignment. Fail: Faculty must manually distribute the submission or the assignment is not recorded.	Supports automated peer review and reduces manual distribution work.
FR-005	Functional	The system shall allow authorized students and faculty evaluators to view the generated test results and itemized rubric score breakdown for a completed submission.	Medium	Pass: An authorized user can open a completed evaluation and view the test results and itemized rubric scores. Fail: The authorized user cannot access the completed evaluation or the results are incomplete.	Allows relevant users to access the results produced by automated evaluation.
NFR-001	Performance & Security	The system shall handle up to 100 concurrent project submissions without dropping queue items or exceeding a 1.5 GB memory footprint.	High	Pass: During a test with 100 concurrent submissions, no queue items are dropped and memory usage remains at or below 1.5 GB. Fail: Any queue item is dropped or memory usage exceeds 1.5 GB.	Ensures reliable operation under the specified concurrent workload.
NFR-002	Performance	The system shall generate and display the itemized rubric score breakdown together with the test results within 10 seconds of a valid submission under normal operating conditions.	High	Pass: At least 95 of 100 valid submissions receive the rubric breakdown and test results within 10 seconds during normal-load testing. Fail: More than 5 of 100 submissions exceed the 10-second limit.	Ensures that evaluation results are provided to users within the specified response time.

UML Use Case Diagram:



Use-Case Flow: Process Project Submission

Use Case ID: UC-02 **Use Case Name:** Evaluate Student Submission

Primary Actor: Faculty Evaluator **Supporting Actor:** Student

Preconditions

1. A student project has been uploaded as a ZIP file.
2. The project has been placed in the evaluation/processing queue.
3. The required unit test suite and evaluation rubric are pre-configured.
4. Rubric configuration is available.

Main Success Scenario

1. The Faculty Evaluator selects a student project from the evaluation queue.
2. The system retrieves the uploaded student project ZIP file.
3. The system executes the pre-configured unit test suite on the submitted project.
4. The system records the results of the executed tests.
5. System executes syntax checks.
6. The system evaluates the submission according to the configured rubric.
7. The system calculates the score for each rubric item.
8. The system generates an itemized rubric score breakdown.
9. The system displays the evaluation results and rubric score breakdown to the Faculty Evaluator.
10. The evaluation is completed successfully.

Alternate Flow – Test or Evaluation Failure

3a. Unit test execution fails

3a1. The system detects that the configured test suite could not be successfully executed or system detects an un-processable ZIP file and the system records the test execution failure.

3a2. The system displays the failure status (error message) to the Faculty Evaluator and the system stops the evaluation.

3a3. The submission is not assigned a valid completed evaluation score.

3a4. The Faculty Evaluator can review the failure and take the appropriate action.

Postconditions

1. The submission has a recorded evaluation status.
2. The test results are recorded by the system.
3. If evaluation is successful, an itemized rubric score breakdown is generated.
4. The evaluation results are available to the Faculty Evaluator